

OPERATING SYSTEMS

Storage Management

S.Thenmozhi

Department of Computer Applications

An allocation method refers to how disk blocks are allocated for files:

1. Contiguous allocation
2. Linked allocation
3. Indexed allocation

Contiguous Allocation

In this scheme, each file occupies a contiguous set of blocks on the disk.

If a file requires n blocks and is given a block b as the starting location, then the blocks assigned to the file will be: $b, b+1, b+2, \dots, b+n-1$.

The directory entry for a file with contiguous allocation contains

- Address of starting block

- Length of the allocated portion.

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directory

file	start	length
count	0	2
tr	14	3
mail	19	6
list	28	4
f	6	2

Advantages

Both the Sequential and Direct Accesses are supported by this.

Extremely fast since the number of seeks are minimal because of contiguous allocation of file blocks.

Disadvantages

This method suffers from both internal and external fragmentation.

Increasing file size is difficult because it depends on the availability of contiguous memory at a particular instance.

Linked Allocation

The disk blocks can be scattered anywhere on the disk.

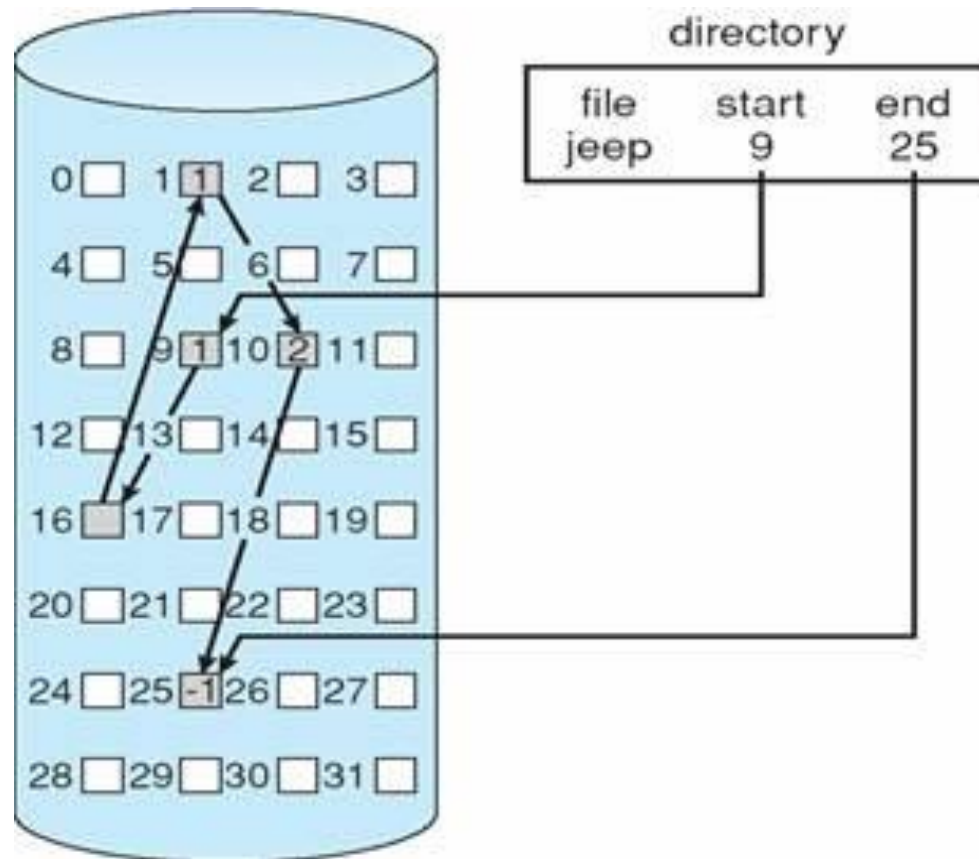
Each file is a linked list of disk blocks which need not be contiguous

The directory entry contains a pointer to the starting and the ending file block.

Each block contains a pointer to the next block occupied by the file.

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Advantages

File size can be increased easily since the system does not have to look for a contiguous chunk of memory.

This method does not suffer from external fragmentation. This makes it relatively better in terms of memory

Disadvantages

The file blocks are distributed randomly on the disk, a large number of seeks are needed to access every block

It does not support random or direct access.

Pointers required in the linked allocation incur some extra overhead.

Indexed Allocation

A special block known as the Index block contains the pointers to all the blocks occupied by a file.

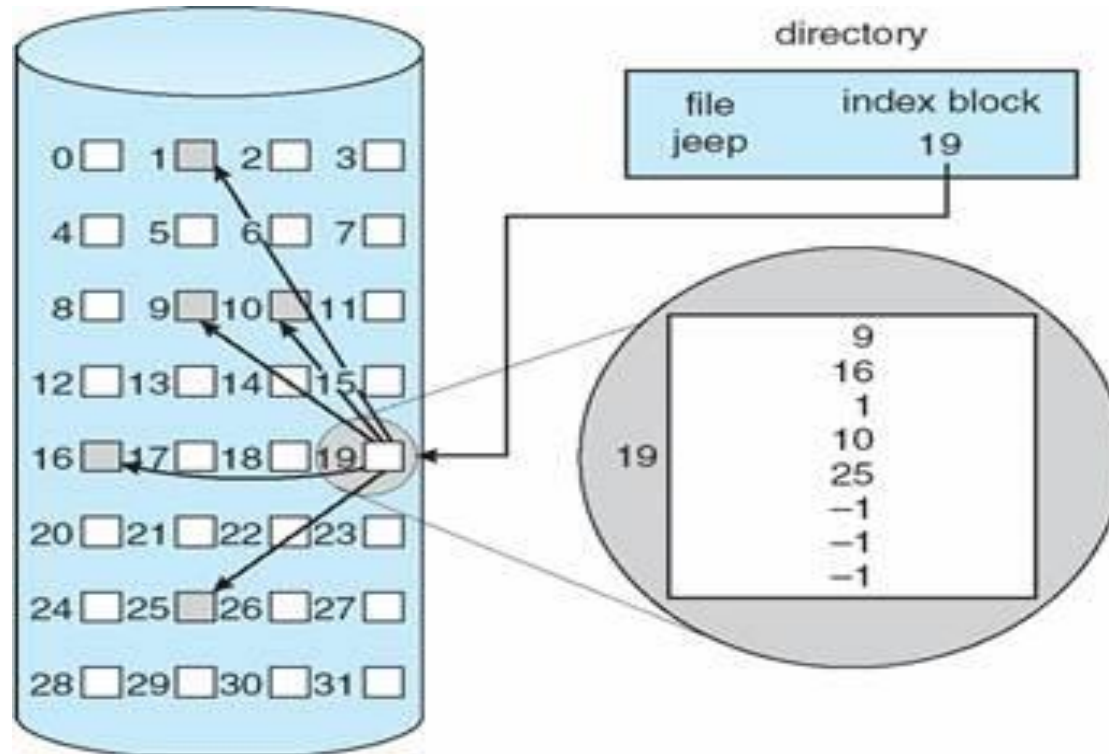
Each file has its own index block.

The i th entry in the index block contains the disk address of the i th file block.

The directory entry contains the address of the index block

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Advantages

This supports direct access to the blocks occupied by the file and therefore provides fast access to the file blocks.

It overcomes the problem of external fragmentation.

Disadvantages

The pointer overhead for indexed allocation is greater than linked allocation.

For files that are very large, single index block may not be able to hold all the pointers.



THANK YOU

S. Thenmozhi

Department of Computer Applications

thenmozhis@pes.edu

+91 80 6666 3333 Extn 393